ECM3412/ECMM409
Nature Inspired Computation
Lecture 7

Swarm Intelligence I:
Ant Colony Optimisation

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- Introduction to Swarm Intelligence
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Swarm Intelligence

Swarms, flocks, etc… often exhibit the following rather interesting properties:

- Individuals of the swarm are incapable of X, or could do X with only low probability.
- However, the *swarm* as a unit is able to do X, with high probability.

The ability to do X is an *emergent property* of the swarm.

Swarm Intelligence II

- Each element of the swarm has its own simple behaviour, and a set of rules for interacting with its fellows, and with the environment.
- Every element is the same – there is no central controller.
- However, X emerges as a result of these local interactions.
- E.g. ants finding food, termites building mounds.

Swarm Algorithms

Inspiration from swarm intelligence has led to some highly successful optimisation algorithms. We will look at:

- Ant Colony Optimisation – a way to solve optimisation problems based on … well, you work it out!
- Particle Swarm Optimisation — a different way to solve optimisation problems, based on the swarming behaviour of several kinds of organisms.

Ant Colony Optimization

My starting point for these slides was some excellent ppt by Tony White (tony@sce.carleton.ca)

With some input from David Corne

Emergent Problem Solving in *Lasius Niger* ants,

For *Lasius Niger* ants, [Franks, 89] observed:

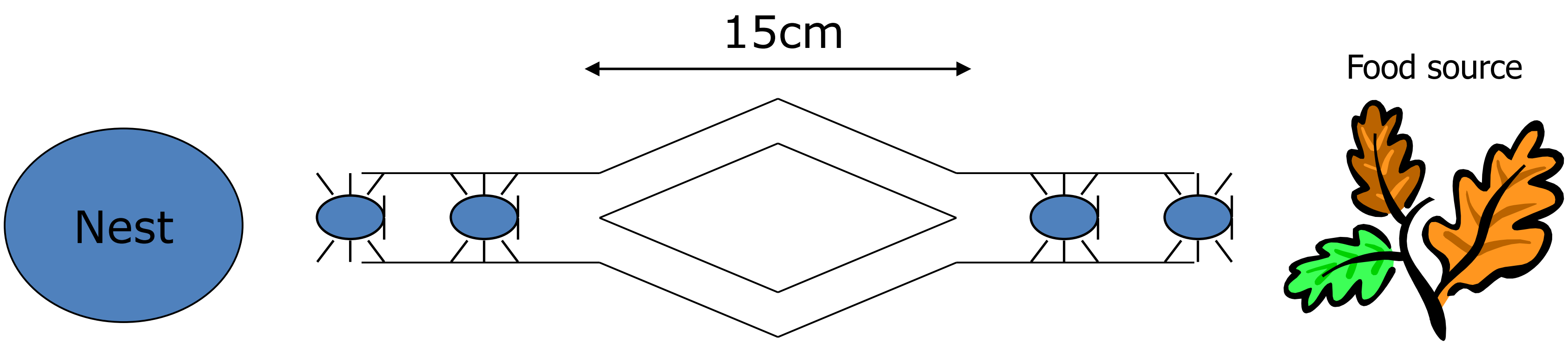
- regulation of nest temperature within 1 degree celsius range;
- forming bridges;
- raiding specific areas for food;
- building and protecting nest;
- sorting brood and food items;
- cooperating in carrying large items;
- emigration of a colony;
- **finding shortest route from nest to food source;**
- preferentially exploiting the richest food source available.

These are swarm behaviours – beyond what any individual can do.

A living bridge

Real Ant Experiments

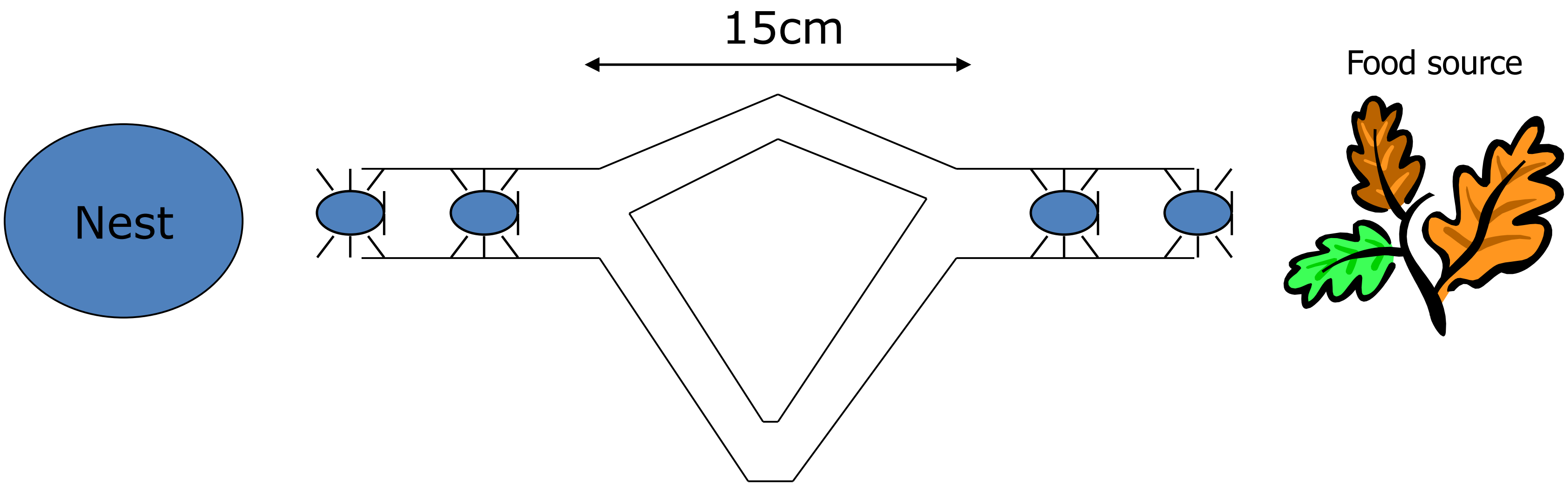
- Experiments conducted on real ants and found very interesting results.
- Deneubourg et al (1989) Double Bridge Experiment



- Ants observed over time
- To begin with - random choices of path
- Later, one path taken by most ants

Real Ant Experiments

- Deneubourg et al (1989) Double Bridge Experiment – 2nd Experiment



- To begin with - random choices of path
- Soon, shortest path selected by most ants
- How?

A key player: Stigmergy

Stigmergy is: indirect communication via interaction with the environment [Gassé, 59]

- Sematonic stigmergy
  - action of agent directly related to problem solving and affects behaviour of other agents (e.g. an ant may orient itself in a certain way to stimulate other ants to begin building a bridge).
- Sign-based stigmergy
  - action of agent affects environment not directly related to problem solving activity.

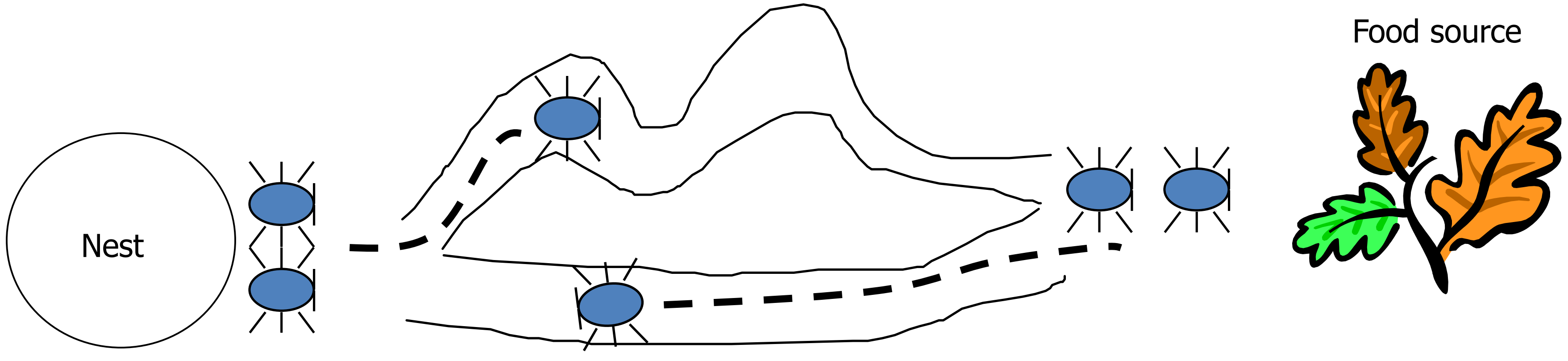
I.e. swarm behaviour emerges from the way individuals communicate through and affect their environment

Ants

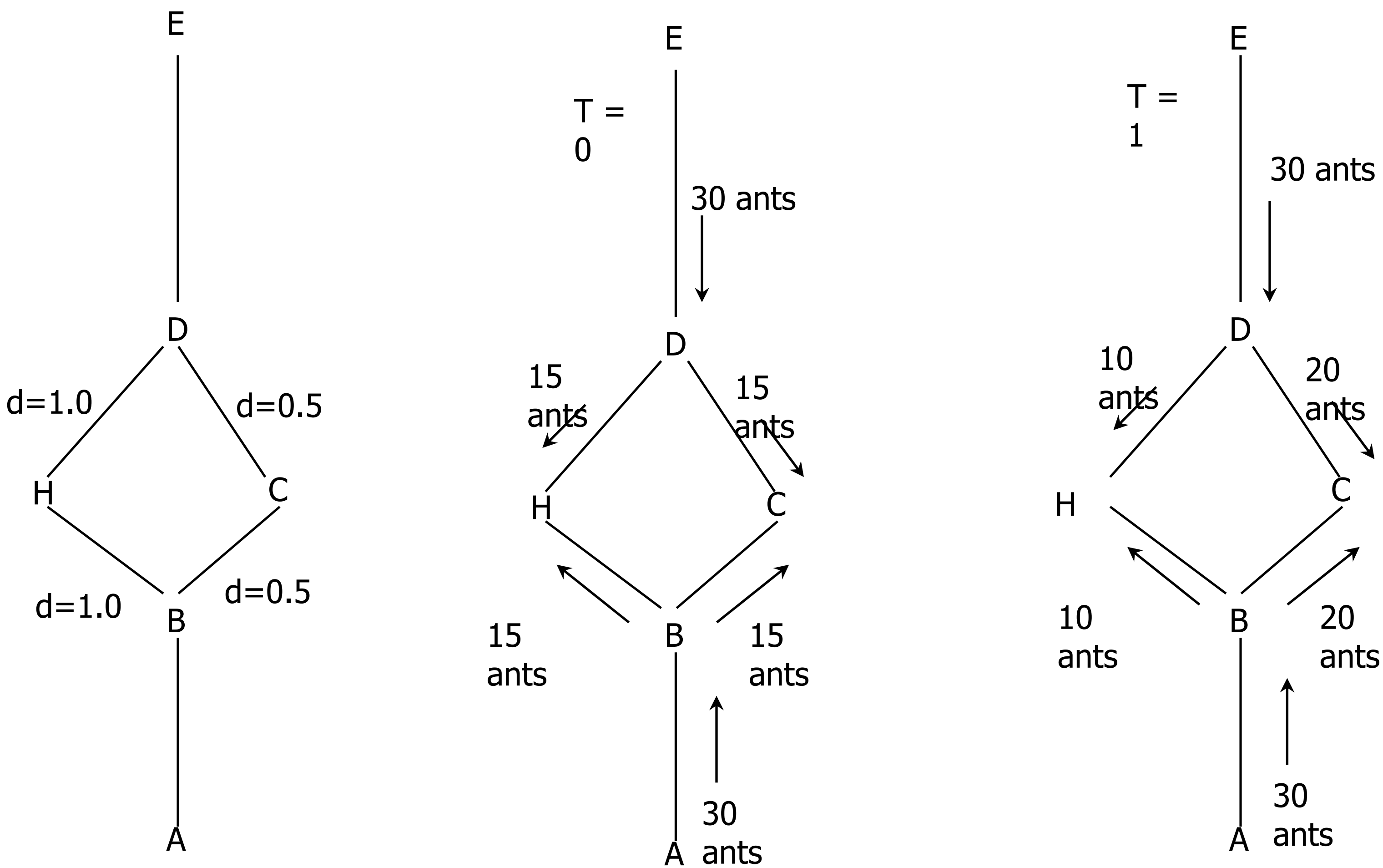
- Ants are behaviorally unsophisticated, but collectively they can perform complex tasks.
- Ants have *highly developed sophisticated sign-based stigmergy*
  - They communicate using pheromones;
  - They lay trails of pheromone that can be followed by other ants.

Pheromone Trails

- Individual ants lay pheromone trails while travelling from the nest, to the nest or possibly in both directions.
- The pheromone trail gradually evaporates over time.
- But pheromone trail strength accumulate with multiple ants using path.
- Ants prefer paths with more pheromone on them.



Basic Operation



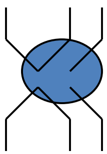
- Time is discrete
- Ants move at 1 unit per timestep
- Example of an autocatalytic process

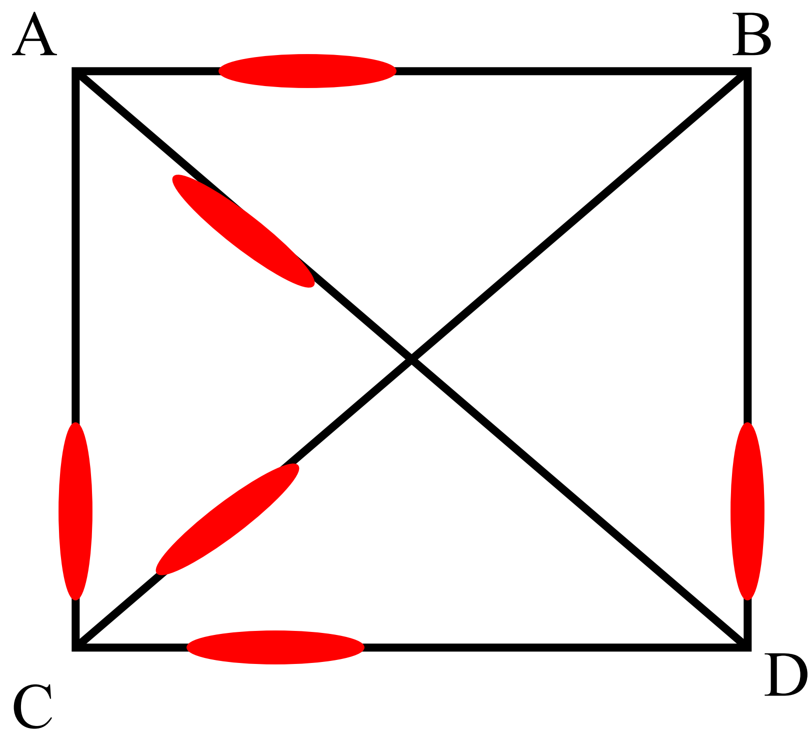
Ant Colony Optimisation Algoirithms: Basic Ideas

- Ants are *agents* that move along edges between nodes in a graph.
- They choose where to go based on pheromone strength (and maybe other things).
- An ant's path represents a specific candidate solution.
- When an ant has finished a solution, pheromone is laid on its path, according to quality of solution.
- This affects behaviour of other ants by `stigmergy' …

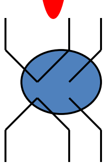
E.g. A 4-city TSP

Initially, random levels of pheromone are scattered on the edges





Pheromone

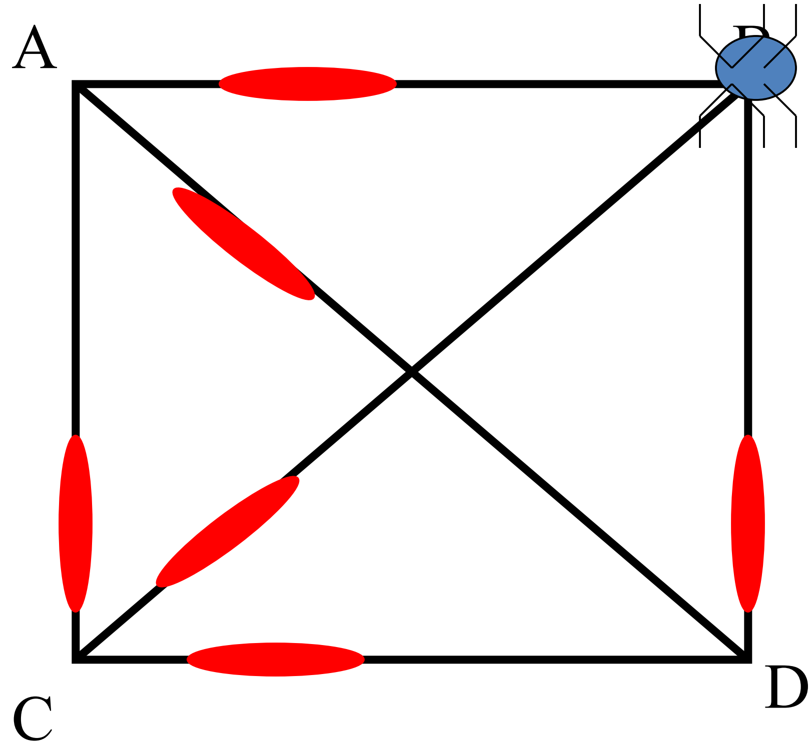


Ant

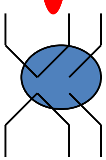
AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

An ant is placed at a random node



Pheromone



Ant

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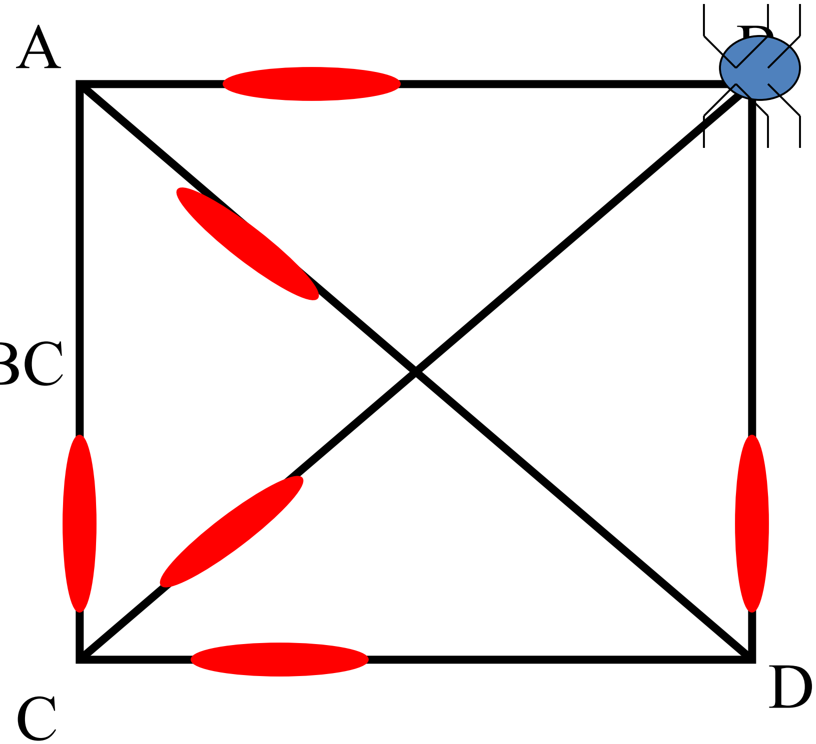
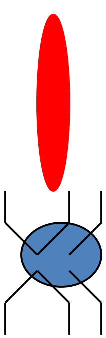
E.g. A 4-city TSP

The ant decides where to go from that node,
based on probabilities
calculated from:
- pheromone strengths,
- next-hop distances.

Suppose this one chooses BC





Pheromone

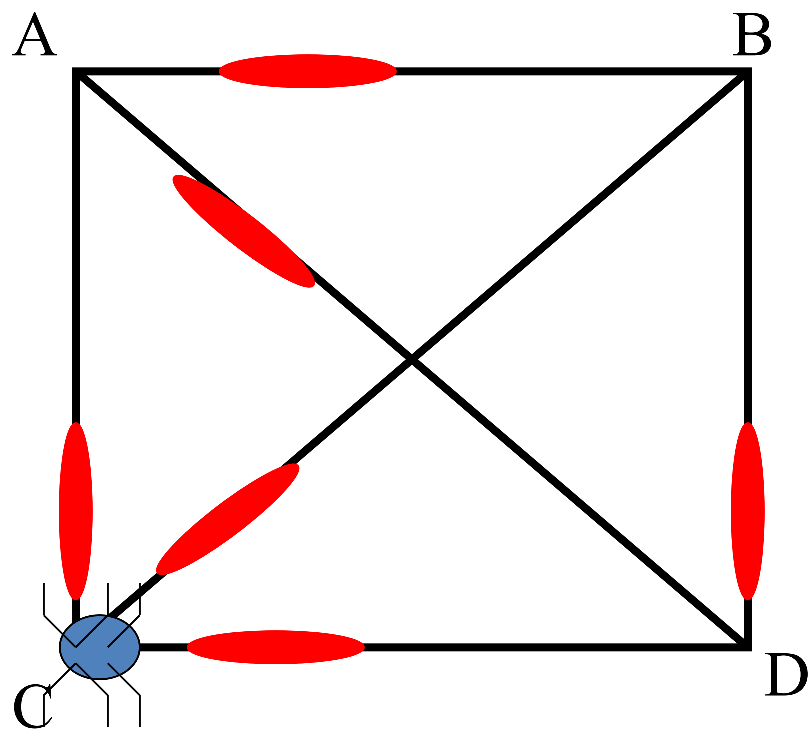
Ant

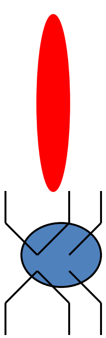
AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

The ant is now at AC, and has a `tour memory' = {B, C} – so he cannot visit B or C again.

Again, he decides next hop (from those allowed) based on pheromone strength and distance; suppose he chooses CD



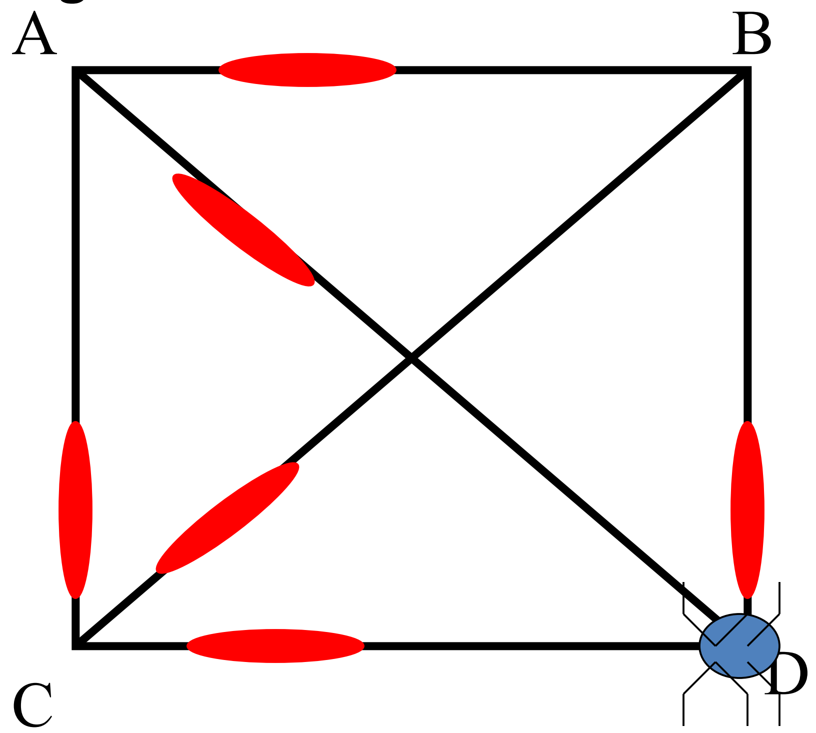


AB: 10, AC: 10, AD, 30, BC, 40, CD 20

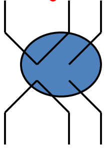
E.g. A 4-city TSP

The ant is now at D, and has a `tour memory' = {B, C, D}

There is only one place he can go now:



Pheromone

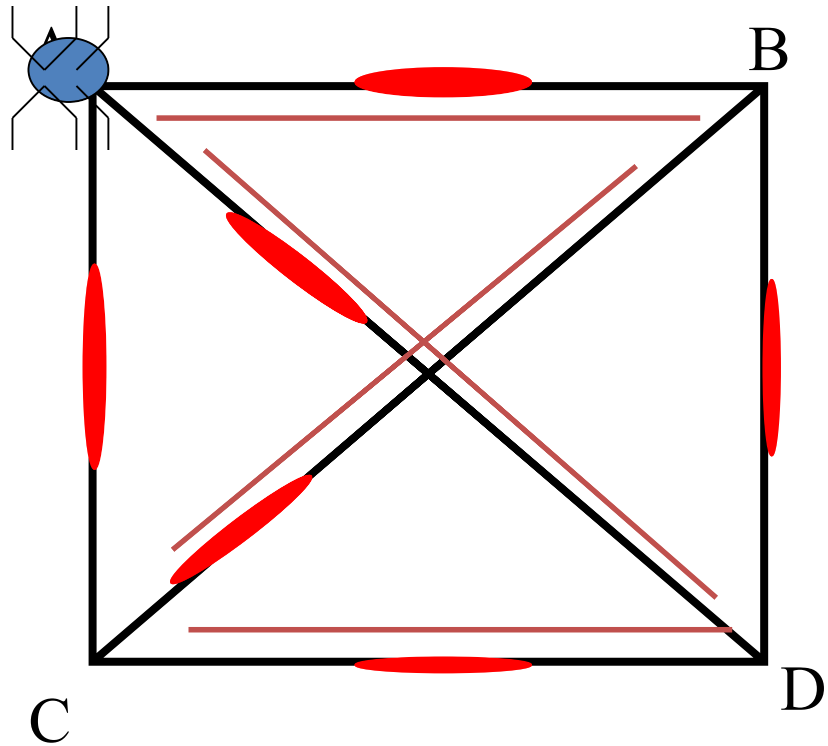


Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

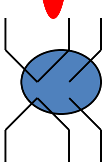
E.g. A 4-city TSP

So, he has finished his tour, having gone over the links:
BC, CD, and DA. AB is added to complete the tour.



Now, evaporate the pheromone on each link.

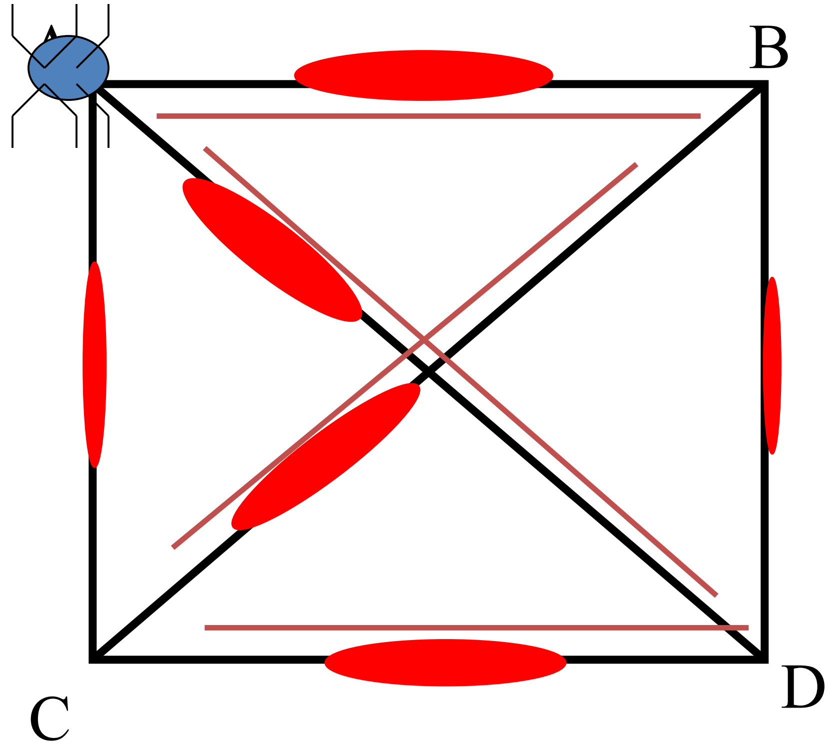

Pheromone

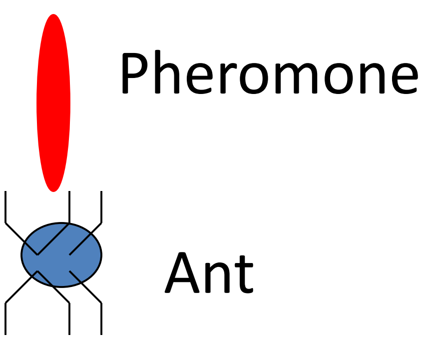

Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

E.g. A 4-city TSP

Next, increase the pheromone on those links that the ant has traversed





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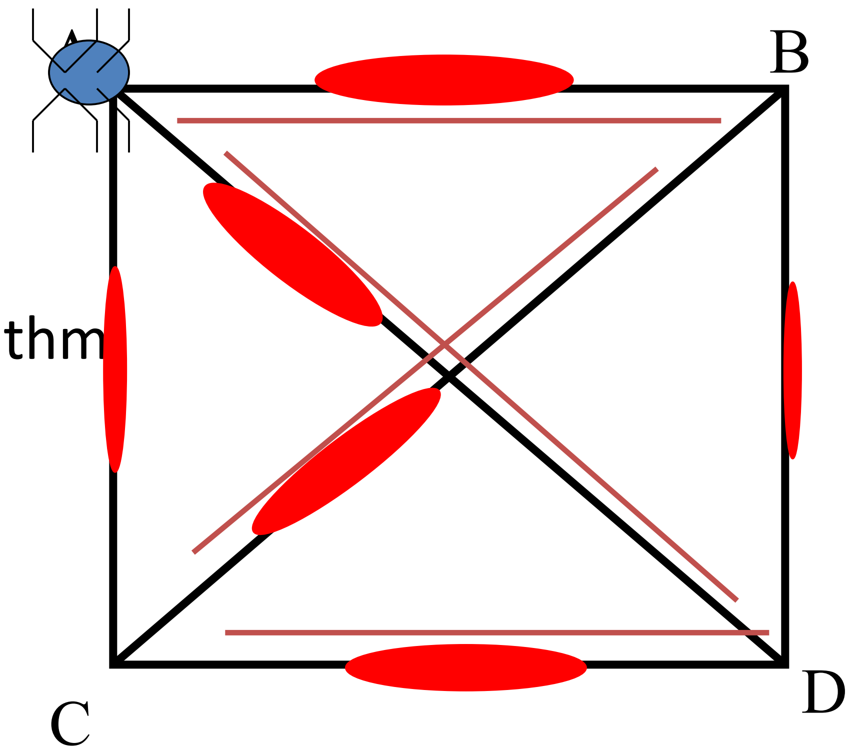
E.g. A 4-city TSP

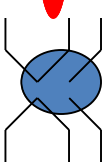
We start again, with another ant in a random position.

Where will he go?

Next time, the actual algorithm and variants.



Pheromone

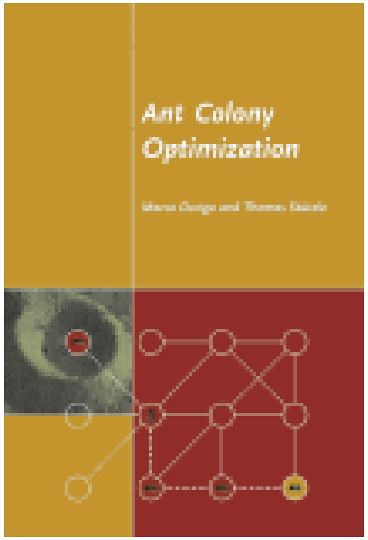


Ant

AB: 10, AC: 10, AD, 30, BC, 40, CD 20

Ant Colony WWW references

- http://iridia.ulb.ac.be/~mdorigo/ACO/ACO.html



Conclusions

- ACO is a powerful technique based on path finding behaviour of real ants
- Relies on the sensing of pheromone of a collective of ants
- Can be used as inspiration to solve many discrete optimisation problems